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Golf green characteristic indicator

Abstract

A golf green characteristic indicator is disclosed. The golf green characteristic indicator may include a ground track and an air track configurable between a travel configuration and an indication configuration. In the travel configuration, the ground track and air track may be detached. In the indication configuration, the ground track is fixated on the golf green, while a golf ball is released from a mechanism on the air track to indicate the golf green characteristic.

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Background/Summary

TECHNICAL FIELD

(1) This disclosure pertains, but not by way of limitation, to the field of indicating characteristics of golf green. More particularly, the disclosure relates to a golf green characteristics indicator for indicating characteristics of the golf green.

BACKGROUND

(2) Characteristics of the golf green of a golf course, such as golf green speed, are measured using a golf stimpmeter. The golf stimpmeter is a linear ramp configured to be positioned inclined at a predefined angle relative to the golf green, from which a golf ball may be iteratively released to the golf green. The average distance traveled by the golf ball from the golf stimpmeter on the golf green is measured (e.g., in feet) and may be utilized to indicate golf green speed. Conventionally, golf stimpmeters have various characteristics, such as repeatable release of a golf ball, misalignment during measuring the golf speed and difficulty in portability.

SUMMARY

(3) A golf green characteristic indicator is disclosed. The golf green characteristic indicator may include a ground track and an air track. The ground track and the air track may be configured between a travel configuration and an indication configuration. In the travel configuration, the ground track may be detached from the air track; in the indication configuration, the ground track may be attached to the air track. The ground track may be fixated on the golf green, and the golf ball may be released from a golf ball release mechanism disposed on the air track to indicate golf green characteristics, i.e., speed of the golf green. The golf green characteristics indication and an indication method for indicating a characteristic of the golf green are explained in detail in successive configurations of this disclosure.

(4) In an illustrative configuration, a golf green characteristic indicator for indicating a characteristic of a golf ball moving on a golf green is disclosed. The golf green characteristic indicator may include a ground track. The ground track may include a proximal end, a distal end oppositely disposed from the proximal end, a first radius of curvature formed at the distal end of the ground track, and a first coupler disposed at the distal end of the ground track. The golf green characteristic indicator may include an air track. The air track may include a proximal end, a distal end oppositely disposed from the proximal end, and a second radius of curvature at the proximal end of the air track. Further, the second radius of curvature is equal to the first radius of curvature of the ground track. The air track may further include a second coupler disposed at the proximal end of the air track. The golf green characteristic indicator may include an alignment pair disposed in either the first coupler or the second coupler. The alignment pair may include a first alignment feature and a second alignment feature. The golf green characteristic indicator may further include a golf ball release mechanism disposed at the distal end of the air track. The golf green characteristic indicator may include a travel configuration. In the travel configuration, the first coupler of the ground track is detached from the second coupler of the air track. The golf green characteristic indicator may further include an indication configuration. In the indication configuration, the first coupler is interfaced with the second coupler, the first alignment feature is aligned with the second alignment feature, the proximal end of the ground track is configured to adjoin the golf green, and the golf ball release mechanism is configured to release the golf ball sequentially along the air track and the ground track with a predetermined release speed as the golf ball exits the proximal end of the ground track to indicate the characteristic of the golf green.

(5) In an illustrative configuration, a golf green characteristic indication method for indicating a characteristic of a golf ball moving on a golf green is disclosed. The golf green characteristic measurement method may include a first step in which a ground track may be provided. The ground track may include a proximal end, a distal end oppositely disposed from the proximal end, a first radius of curvature formed at the distal end of the ground track, and a first coupler disposed at the distal end of the ground track. In the next step, an air track may be provided. The air track may include a proximal end, a distal end oppositely disposed from the proximal end, and a second radius of curvature at the proximal end of the air track. Further, the second radius of curvature is equal to the first radius of curvature of the ground track. The air track may further include a second coupler disposed at the proximal end of the air track. In the next step, an alignment pair disposed in either the first coupler or the second coupler may be provided. The alignment pair may include a first alignment feature and a second alignment feature. In the next step, a golf ball release

mechanism disposed at the distal end of the air track may be provided. In the next step, the ground track and the air track may be configured between a travel configuration and an indication configuration. The travel configuration may include detaching the first coupler of the ground track from the second coupler of the air track. The indication configuration may include interfacing the first coupler with the second coupler, aligning the first alignment feature with the second alignment feature, the proximal end of the ground track is configured to adjoin the golf green, and releasing the golf ball release mechanism. Further, the golf ball release mechanism, on releasing, is configured to release the golf ball sequentially along the air track and the ground track with a predetermined release speed as the golf ball exits the proximal end of the ground track for indicating the characteristic of the golf green.
(6) Further areas of applicability of the present disclosure will become apparent from the detailed description provided hereinafter. It should be understood that the detailed description and specific examples while indicating various configurations, are intended for purposes of illustration only and are not intended to necessarily limit the scope of the disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The accompanying figures of the drawing, which are included to provide a further understanding of general aspects of the system/method, are incorporated in and constitute a part of this specification. These illustrative aspects of the system/method, together with the detailed description, explain the principles of the system. No attempt is made to show structural details in more detail than necessary for a fundamental understanding of the system and the various ways it is practiced. The following figures of the drawing include:
(2) FIG. **1** illustrates a schematic of an individual/person determining a golf green characteristic with a golf green characteristic indicator (in an indication configuration);
(3) FIG. **2** illustrates a schematic of a golf green characteristic indicator (in a travel configuration);
(4) FIG. **3** illustrates a perspective view of a golf green characteristic indicator;
(5) FIG. **4** illustrates a left-side view of a golf green characteristic indicator;
(6) FIG. **5** illustrates a right-side view of a golf green characteristic indicator;
(7) FIG. **6** illustrates an exploded view of the golf green characteristic indicator;
(8) FIG. **7** illustrates a perspective view of the golf ball release mechanism;
(9) FIG. **8** illustrates a schematic illustratively depicting the golf ball traveling through the air track and the ground track;
(10) FIG. **9** illustrates a schematic illustratively depicting the golf ball traveling through the golf green; and
(11) FIG. **10** illustrates a flowchart of a golf green characteristic indication method for indicating a characteristic of a golf green.
(12) In the appended figures, similar components and/or features may have the same reference label. Further, various components of the same type may be distinguished by following the reference label with a dash and a second label that distinguishes among the similar components. If only the first reference label is used in the specification, the description is applicable to any one of the similar components having the same first reference label, irrespective of the second reference label. Where the reference label is used in the specification, the description is applicable to any one of the similar components having the same reference label.
DETAILED DESCRIPTION
(13) Illustrative configurations are described with reference to the accompanying drawings. Wherever convenient, the same reference numbers are used throughout the drawings to refer to the same or like parts. While examples and features of disclosed principles are described herein,

modifications, adaptations, and other implementations are possible without departing from the spirit and scope of the disclosed configurations. It is intended that the following detailed description be considered exemplary only, with the true scope and spirit being indicated by the following claims.
(14) Conventional golf stimpmeters can be relatively large and heavy, making them less portable. While smaller, more compact versions may be available, such golf stimpmeters may not always provide the same level of precision as larger models. Moreover, suppose the stimpmeter is not placed in the correct position (e.g., not on a completely flat surface or misaligned with the slope of the green). In that case, inaccurate characteristics or measurements related to green speed may be recorded.
(15) In an effort to make the golf stimpmeter portable and enable accurate measurement of characteristics of the golf green, the present disclosure relates to a golf green characteristic indicator. The golf green characteristic indicator may include a ground track, an air track, and a golf ball release mechanism. The golf ball release mechanism may be configured to release a golf ball, thereby allowing the golf ball to move towards the golf green via the air track and the ground track. The golf green characteristic indicator may be configured in an indication configuration and a traveling configuration. The present disclosure explains the golf green characteristic indicator and configurations in detail, in conjunction with FIGS. **1-10**.
(16) FIG. **1** illustrates a schematic **100** of an individual/person determining a golf green characteristic with a golf green characteristic indicator **102** in an indication configuration. In the indication configuration, a user may position the golf green characteristic indicator **102** on the golf green **104**. In some configurations, the user may release the golf ball **106** from the golf green characteristic indicator **102** towards the golf green **104**. Each golf ball **106**, after being dispatched from the golf green characteristic indicator **102**, may be configured to travel a predefined distance "D" (ft) on the golf green **104**. Such process is iterated to determine one or more values of "D"; hence, the average value of "D" corresponds to the golf green characteristic, i.e., golf green speed of the golf green **104**.
(17) FIG. **2** illustrates schematic **200** of the individual carrying the golf green characteristic indicator **102** of FIG. **1** in the travel configuration. The golf green characteristic indicator **102**, after measuring the golf green characteristic of the golf green **104**, may be disassembled and stacked within a travel bag **202**. Accordingly, the golf green characteristic indicator **102** may be accommodated within the travel bag **202**, providing ease of portability. The golf green characteristic indicator **102** is explained in detail hereinafter.
(18) FIG. **3** illustrates a perspective view **300** of a golf green characteristic indicator **102**. The golf green characteristic indicator **102** may include a ground track **302** and an air track **304**. The golf green characteristic indicator **102** may include a golf ball release mechanism **306** positioned on the air track **304**. Further, the ground track **302** may include a ground track guide **310**, and the air track **304** may include an air track guide **312**. Moreover, in the indication configuration, the golf green characteristic indicator **102**, may be mounted on a stand **308** adjoined to the ground track **302**.
(19) The ground track **302** may be configured to couple with the air track **304**. As such, in some configurations, when the ground track **302** may be coupled to the air track **304**, the ground track guide **310** and the air track guide **312** may be aligned to form a continuous track guide. The continuous track guide may be configured to guide the golf ball **106** released from the golf ball release mechanism **306** towards the golf green **104**.
(20) FIG. **4** illustrates a left-side view **400** of a golf green characteristic indicator **102**. The ground track **302** may include a ground track distal end **402** and a ground track proximal end **404** disposed oppositely to the ground track distal end **402**. The ground track proximal end **404** may be positioned on the golf green **104**. Further, the stand **308** may be coupled to the ground track distal end **402**. Moreover, the ground track proximal end **404** may include a base **406** and a recess **408**. The base **406** may include a planar bottom, which may conform with the surface of the golf green

104. The recess **408** may be formed on the planar bottom of the golf green **104** and may be configured to accommodate the stand **308** in the travel configuration.

(21) The air track **304** may include an air track proximal end **410** and an air track distal end **412**. The air track proximal end **410** may be coupled to the ground track distal end **402** to couple the ground track **302** and the air track **304**. As such, in some configurations, the ground track **302** and the air track **304** may collectively form a continuous curved profile, such as a parabolic-shaped curve. The parabolic-shaped curve, in addition to the continuous guide track, may ensure seamless and uninterrupted travel of the golf ball **106** from the air track **304** to the golf green **104**. The parabolic-shaped curve collectively formed by the ground track **302** and the air track **304** is explained in detail hereinafter.

(22) FIG. **5** illustrates a right-side view **500** of a golf green characteristic indicator **102**. As explained earlier, the ground track **302** and the air track **304** may collectively form a continuous curved profile to allow seamless and uninterrupted travel of the golf ball **106** from the air track **304** to the golf green **104**. The continuous curved profile is defined as a parabolic-shaped curve, with a variable radius of curvature across each point on the ground track **302** and the air track **304**. For example, the radius of curvature from an imaginary center "O" may vary across the ground track **302** and the air track **304**.

(23) With continued reference to FIG. **5**, the radius of curvature at the golf ball release mechanism **306**, or the air track distal end **412** is denoted by "R.sub.1", the radius of curvature at the air track proximal end **410** is denoted by "R.sub.2", the radius of curvature at the ground track distal end **402** is denoted by "R.sub.3", and the radius of curvature at the ground track proximal end **404** is denoted by "R.sub.4". As seen, the R.sub.1, R.sub.2, R.sub.3, and R.sub.4 may be measured from the center "O." As such, in some configurations, the radius of curvatures may increase from the golf ball release mechanism **306** towards the ground track proximal end **404**. For example, the magnitude of R.sub.1 may be less than the magnitude of R.sub.2, and the magnitude of R.sub.3 may be less than the magnitude of R.sub.4. This increase in the radius of curvature, from R.sub.1 to R.sub.4 may result in the shaping of the continuous curved profile formed by the ground track **302** and the air track **304** as the parabolic-shaped curve.

(24) To maintain consistency of the curve, the magnitude of R.sub.2 may be equal to the magnitude of R.sub.3. Put differently, the radius of curvature at the ground track proximal end **404** may be equal to the radius of curvature at the air track proximal end **410**. Such configuration may allow the formation of a smooth track or smooth contour of the continuous track guide between the air track distal end **412** and the ground track distal end **402**. Thus, the movement of golf ball **106** may be uninterrupted on both the ground track **302** and the air track **304** upon being released by the golf ball release mechanism **306**. Advantageously, losses in velocity of the golf ball **106** incurred due to such interruption may be prevented.

(25) With continued reference to FIG. **5**, the difference in magnitude of R.sub.1 and R.sub.3 may result in the air track distal end **412** and the ground track distal end **402** formed at various inclinations. For instance, as a result of R.sub.1, the air track distal end **412** may be configured at an inclination θ.sub.1 varying from about 30° to about 40°. Further attributable to R.sub.4, the ground track distal end **402** may be configured at an inclination θ.sub.2 ranging from about 20° to about 25°. As such, the inclination θ.sub.2 of the ground track distal end **402** may be equivalent to an exit angle or an angle at which the conventional golf stimpmeters are required to be inclined to release the golf ball **106** to the golf green **104**.

(26) With continued reference to FIG. **5**, the golf green characteristic indicator **102** may be formed of a height H ranging from about 8 inches to about 10 inches or from about 0.6 ft to about 1 ft. Moreover, the golf green characteristic indicator **102** may be formed of a length L from about 16 inches to about 18 inches, or from about 1.4 ft to about 1.8 ft. Accordingly, with such dimensions and with inclinations θ.sub.1, θ.sub.2, the golf ball **106** may exit at the requisite velocity to measure the characteristics of the golf green **104**.

(27) FIG. **6** illustrates an exploded view **600** of the golf green characteristic indicator **102**. As explained earlier, the golf green characteristic indicator **102** may include the golf ball release mechanism **306**. The golf ball release mechanism **306** may include a support plate **602** adjoined to the air track distal end **412**. The support plate **602** may include a left face **604**, a right face **606**, a central hole **608**, a first torsion spring pocket **610**, and a radial pocket **612**. Further, the golf ball release mechanism **306** may include a golf ball cradle **614**. The golf ball cradle **614** may include a central shaft **616**. Further, the golf ball cradle **614** may include a second torsion spring pocket **618**. The golf ball cradle **614** may be shaped as a hollow cylindrical segment configured to accommodate the golf ball **106** prior to release for measuring characteristics of the golf green **104**.
(28) The golf ball release mechanism **306** may further include a knob **620** coupled to the golf ball cradle **614**. For example, the knob **620** may be coupled to the golf ball cradle **614** via a pin **622**. To further elaborate, the pin **622** may be configured to pass through the radial pocket **612** and engage to a slot **624** formed on the golf ball cradle **614**. Simultaneously, the pin may be coupled to the knob **620**, thereby coupling the knob **620** to the golf ball cradle **614**. In addition, the knob **620** may also be coupled to the central shaft **616**. Henceforth, the rotation of knob **620** may simultaneously rotate the golf ball cradle **614** to release the golf ball **106**.
(29) The golf ball release mechanism **306** may further include a torsion spring **626** biasingly adjoining the support plate **602** and the golf ball cradle **614**. The torsion spring may include a coil **628**, a first leg **630**, and a second leg **632**. The first leg **630** may extend from the coil **628**, and the second leg **632** may further extend from the coil **628** in a direction opposite to the first leg **630**. Moreover, the first leg **630** may be adjoined to the first torsion spring pocket **610**, and the second leg **632** may be coupled to the knob **620**. Further, the coil **628** may be disposed in the central hole **608** and coupled to the central hole **608**. The coil **628** may be wound about the central shaft **616** and coupled to the second torsion spring pocket **618**. The knob **620** may be rotated against a residual torsion force in the torsion spring **626** from an original position in a counterclockwise direction to rotate the golf ball cradle **614**. After rotation, the knob **620** may be reinstated to the original position by the residual torsion force in the torsion spring **626**.
(30) When assembled, the knob **620** may adjoin the left face **604** of the support plate **602**, and the golf ball cradle **614** may adjoin the right face **606** of the support plate **602**. Moreover, a friction reduction element **634**, such as a bushing ring, may be disposed between the golf ball cradle **614** and the right face **606** of the support plate **602**. The friction reduction element **634** may reduce friction between the golf ball cradle **614** and the right face **606** during rotation of the golf ball cradle **614**.
(31) With continued reference to FIG. **6**, the golf green characteristic indicator **102** may include a first coupler **636** and a second coupler **638**. The first coupler **636** may be formed in the ground track distal end **402**, and the second coupler **638** may be formed in the air track proximal end **410**. The first coupler **636** may be coupled to the second coupler **638** to align and couple the ground track **302** and the air track **304**. To further elaborate, each of the first coupler **636** and the second coupler **638** may include an alignment pair and a magnet pair configured to align and couple the ground track distal end **402** with the air track proximal end **410** in the indication configuration.
(32) The alignment pair may include a first alignment feature **640** and a second alignment feature **642**. The first alignment feature **640** may be formed in the first coupler **636**, and the second alignment feature **642** may be formed in the second coupler **638**. Further, the first alignment feature **640** may be slidingly engaged with the second alignment feature **642** to align the first coupler **636** and the second coupler **638** in the indication condition. In some configurations, the first alignment feature **640** may be designed in conformity with the second alignment feature **642** to ensure a proper alignment of the first coupler **636** and the second coupler **638**, thereby aligning the ground track **302** with the air track **304**. In an illustrative configuration, any one of the first alignment feature **640** or the second alignment feature **642** may include a tapered protrusion, and either of the first alignment feature **640** or the second alignment feature **642** may include a tapered pocket. For

example, if the first alignment feature **640** may be formed as a tapered protrusion, the second alignment feature **642** may be formed as a tapered pocket, and vice versa.

(33) Accordingly, the interfacing of the first alignment feature **640** with the second alignment feature **642** may result in the accommodation of the tapered protrusion within the tapered pocket. Due to the tapered shape of the tapered protrusion and the tapered shape of the tapered pocket in conformity with the tapered protrusion, the first coupler **636** may be aligned, and locked with the second coupler **638**. Such alignment prevents rotation or movement of the first coupler **636** relative to the second coupler **638**, thereby preventing the rotation of the ground track **302** relative to the air track **304**.

(34) With continued reference to FIG. **6**, the alignment and coupling of the first alignment feature **640** with the second alignment feature **642** may be reinforced with the magnetic pair. In addition to the alignment features, the magnetic pair may be disposed within the first coupler **636** and the second coupler **638**. The magnetic pair may include a first magnet **644** of a first magnetic polarity and a second magnet **646** of a second magnetic polarity. The second polarity may be opposite to the first polarity. Alternatively, the second magnet **646** may be replaced by a ferromagnetic ring, which may be configured to be attracted to the first magnet **644**. The first magnet **644** may be disposed in either the first coupler **636** or the second coupler **638**, and the second magnet **646** may be disposed in either the first coupler **636** or the second coupler **638**.

(35) Particularly, the first magnet **644** may be disposed in a first pair of magnet pockets **648** formed in either of the first coupler **636** or the second coupler **638**. Similarly, the second magnet **646** may be disposed in a second pair of magnet pockets **650** formed in either the first coupler **636** or the second coupler **638**. As seen in FIG. **6**, the first magnet **644** may be disposed and affixed within a first pair of magnet pockets **648** (not shown in the figure) formed in the first coupler **636** using magnet fasteners **652**. Moreover, the second magnet **646** may be disposed in a second pair of magnet pockets **650** formed in the second coupler **638** using magnet fasteners **654**.

(36) As may be appreciated, the first magnet **644** may attract the second magnet **646** and form a magnetic coupling. Accordingly, the magnetic coupling may be configured to couple the first coupler **636** and the second coupler **638**, in addition to the alignment implemented using the first alignment feature with the second alignment feature. Accordingly, with the coupling of the first coupler **636** with the second coupler **638** using the magnetic pair and the alignment pair, the ground track **302** may be coupled to the air track **304** with a proper consistency of the continuous curved profile.

(37) With continued reference to FIG. **6**, the second coupler **638** may include a plurality of stand pockets **656**. The plurality of stand pockets **656** may be coupled to the stand **308**. Particularly, the stand **308** may include a pair of legs **658** rotatably coupled to the plurality of stand pockets **656**. Moreover, the stand **308** may include a web **660** formed between the pair of legs **658**. The web **660** may be coupled to the recess **408** in the travel configuration.

(38) The plurality of stand pockets **656** may be configured to constrain rotation of the pair of legs **658** up to a predefined angle relative to the golf green **104**. In the indication configuration, the stand **308** or the web **660** may be rotationally decoupled from the recess **408** and may be rotated by a predefined inclination angle relative to the ground track distal end **402** or the second coupler **638**. Hence, the stand **308** may be adjoined to the golf green **104** at a predefined inclination angle to align the ground track proximal end **404** with the golf green **104** and provide an appropriate elevation to the air track distal end **412**. After measuring the golf green characteristics in the travel configuration, the stand **308** may be rotatably coupled to the recess **408**.

(39) The ground track proximal end **404** may include a predetermined list of features **662**. The predetermined list of features **662** may include a hook **664**, a transverse slot **666**, and a plurality of golf tee slots **668**. The predetermined list of features **662** may enable a proper alignment of the ground track proximal end **404** with the golf green **104** and enable a proper measurement of the characteristics of the golf green **104**. For example, the plurality of golf tee slots **668** may enable

passage of the plurality of golf tee (refer to a plurality of golf tees **802** in FIG. **8**) through or put differently, the golf tee is inserted in the plurality of golf tee slots **668** as the ground track distal end **402** may be positioned on the golf green **104**. As such, the plurality of the golf tee may be configured to nail the ground track distal end **402** on the golf green **104**, thereby preventing misalignment of the golf green characteristic indicator **102** while measuring characteristics of the golf green **104**. Moreover, the transverse slot **666** may be configured to receive an end hook of a tape measure or a measuring tape. Similarly, the hook **664** may be configured to engage with a loop of a ruler. As may be appreciated, the tape measure or the ruler may be utilized to measure the distance traveled by the golf ball **106** after exiting the ground track **302**.
(40) FIG. **7** illustrates a perspective view **700** of the golf ball release mechanism **306**. In the indication configuration, the golf ball release mechanism **306**, as explained earlier, may be configured to release the golf ball **106** from the air track distal end **412**. Particularly, the golf ball cradle **614** may be configured to accommodate the golf ball **106**. As explained earlier, the golf ball cradle **614** may be formed as a hollow cylindrical segment. The hollow cylindrical segment may be formed of a predefined radius measured from the central shaft **616** and a length similar to the width of the air track distal end **412**.
(41) The hollow cylindrical segment may be configured to accommodate the golf ball **106**. When the golf ball cradle **614** may be rotated using the knob **620**, the hollow cylindrical segment may rotate in a circular trajectory similar to a circumference of an imaginary cylinder formed of a radius similar to the radius of the hollow cylindrical segment. Accordingly, the golf ball **106** may be dropped on the air track distal end **412**. Subsequently, the golf ball **106** may travel along the air track **304** and the ground track **302**.
(42) FIG. **8** illustrates a schematic **800** illustratively depicting the golf ball **106** traveling through the air track **304** and the ground track **302**. Initially, in the indication configuration, the first coupler **636** may be interfaced with the second coupler **638**, and the first alignment feature **640** may be aligned with the second alignment feature **642** to couple the ground track **302** and the air track **304**. Further, the ground track proximal end **404** may be configured to adjoin the golf green **104**. Moreover, the ground track proximal end **404** may be aligned and nailed in the golf green **104** using a plurality of golf tees **802**. Further, the golf ball release mechanism **306** may be configured to release the golf ball **106** sequentially along the ground track **302** and the air track **304**.
(43) The golf ball **106** may be configured to travel along ground track **302** and air track **304** at a predetermined speed. The predetermined speed may depend on an angle of inclination and an elevation at which the golf ball **106** may be released, or the elevation of the air track distal end **412**. The predetermined speed is calculated using an energy conversion equation:
mgH=½*mv*.sub.h.sup.2   (1)
where, m=mass of the golf ball;
(44) g = accelerationduetogravity9.81$(\frac{m}{s^2})$or32.2$(\frac{\text{ft}}{s^2})$; H=elevation of the air track distal end **412** relative to the golf green **104**; and v.sub.h=v cos θ.sub.2, a predetermined release speed of the golf ball **106**. where, θ.sub.2=an inclination angle of the golf green characteristic indicator **102**, or angle of inclination of the ground track proximal end **404** relative to the golf green **104**.
(45) In an exemplary configuration, the mass (m) of the golf ball **106** may be about 46 grams, and the elevation (H) of about 1 ft, and with the angle of inclination θ.sub.2 of the ground track proximal end **404** relative to the golf green **104** ranging about 20°. With equation (1), the predetermined release speed of the golf ball **106** may be obtained from about 6 ft/second to about 8 ft/second. Such speed of the golf ball **106** may be the requisite speed to exit the ground track **302** and allow travel of the golf ball **106** along the golf green **104**.
(46) FIG. **9** illustrates a schematic **900** illustratively depicting the golf ball **106** traveling through the golf green **104**. The travel of the golf ball **106** along the golf green **104** may be measured using a tape measure **902** or a ruler (not shown). For instance, a hook **904** of the tape measure **902** may be received by the transverse slot **666**, and the tape may be extended to a point at which the motion

of the golf ball **106** may be ceased. Accordingly, the distance from the ground track proximal end **404** and the point at which the golf ball **106** may be stopped may be calculated as D (in feet). The distance D may be iteratively measured, and an average of the magnitude of D may be referred to as a characteristic of the golf green **104**, or in other words, the speed of the golf green **104**.
(47) FIG. **10** illustrates a flowchart **1000** of a golf green characteristic indication method for indicating a characteristic of a golf green **104**. The golf green characteristic method may include a plurality of steps to be implemented to assemble the golf green characteristic indicator **102** and determine the characteristic of the golf green **104**, or in other words, determine the speed of the golf green **104**. The steps of the golf green characteristic method are explained hereinafter.
(48) At step **1002**, a ground track **302** may be provided. The ground track **302** may include ground track distal end **402** and a ground track proximal end **404** oppositely disposed to the ground track distal end **402**. Further, the ground track distal end **402** may be formed at a radius of curvature R.sub.2. Moreover, the ground track **302** may include a first coupler **636** disposed at the ground track distal end **402**.
(49) At step **1004**, an air track **304** may be provided. The air track **304** may include an air track proximal end **410** and an air track distal end **412** oppositely formed at the air track proximal end **410**. Further, the air track proximal end **410** may be formed at a radius of curvature R.sub.1. Moreover, the air track **304** may include a second coupler **638** disposed at the air track proximal end **410**. As such, in some configurations, the radius of curvature R.sub.1 may be equal to the radius of curvature R.sub.2.
(50) At step **1006**, an alignment pair may be provided. The alignment pair may include a first alignment feature **640** and a second alignment feature **642**. The first alignment feature **640** may be formed in the first coupler **636**, and the second alignment feature **642** may be formed in the second coupler **638**. The first alignment feature **640** may be slidingly engaged with the second alignment feature **642** to align the first coupler **636** and the second coupler **638**.
(51) At step **1008**, a golf ball release mechanism **306** may be provided. Particularly, the golf ball release mechanism **306** may be adjoined to the air track distal end **412**. Further, the golf ball release mechanism **306** may include a support plate **602** adjoined to the air track distal end **412**. The support plate **602** may include a left face **604**, a right face **606**, a central hole **608**, a first torsion spring pocket **610**, and a radial pocket **612**. Further, the golf ball release mechanism **306** may include a golf ball cradle **614**. The golf ball cradle **614** may include a central shaft **616**. Further, the golf ball cradle **614** may include a second torsion spring pocket **618**. The golf ball cradle **614** may be shaped as a hollow cylindrical segment configured to accommodate the golf ball **106** prior to release for measuring characteristics of the golf green **104**. Further, the golf ball release mechanism **306** may include a knob **620**. The knob **620** may also be coupled to the central shaft **616**. As such, the rotation of knob **620** may simultaneously rotate the golf ball cradle **614** to release the golf ball **106**.
(52) At step **1010**, the golf green characteristic indicator **102**, particularly the ground track **302** and the air track **304** may be configured between a travel configuration and an indication configuration. In the indication configuration, the first coupler **636** may be interfaced with the second coupler **638**, and the first alignment feature **640** may be aligned with the second alignment feature **642** to couple the ground track **302** and the air track **304**. The ground track **302** and the air track **304** may collectively form a continuous curved profile. Further, the ground track proximal end **404** may be configured to adjoin and nailed in the golf green **104** using a plurality of golf tees **802**. Further, the golf ball release mechanism **306** may be actuated to release the golf ball **106** sequentially along the ground track **302** and the air track **304** to indicate or measure the characteristic of the golf green **104**. After the indication of the golf green **104**, the golf green characteristic indicator **102** may be configured in the travel configuration. In the travel configuration, the first coupler **636** may be detached from the second coupler **638**, separating the ground track **302** and the air track **304**.
(53) The methods, systems, devices, graphs, and/or tables are illustrative examples, and

configurations may omit, substitute, or add various procedures or components as appropriate. For instance, the methods may be reordered in alternative configurations, and/or various stages may be added, omitted, and/or combined. Alternatively, features described with respect to certain configurations may be in various alternative configurations. Different aspects and elements of the configurations may be combined similarly. Also, technology evolves; thus, many of the elements are examples and do not limit the scope of the disclosure or claims. Additionally, the techniques discussed herein may provide differing results with different types of context awareness classifiers.
(54) Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly or conventionally understood. As used herein, the articles “a” and “an” refer to one or more than one (i.e., to at least one) of the grammatical object of the article. By way of example, “an element” means one element or more than one element. “About” and/or “approximately” as used herein when referring to a measurable value such as an amount, a temporal duration, and the like encompass variations of ±20% or ±10%, ±5%, or ±0.1% from the specified value as such variations are appropriate in the context of the systems, devices, circuits, methods, and other implementations described herein. “Substantially,” as used herein when referring to a measurable value such as an amount, a temporal duration, a physical attribute (such as frequency), and the like, also encompasses variations of ±20% or ±10%, ±5%, or ±0.1% from the specified value as such variations are appropriate in the context of the systems, devices, circuits, methods, and other implementations described herein.
(55) As used herein, including in the claims, “and” as used in a list of items prefaced by “at least one of” or “one or more of” indicates that any combination of the listed items may be utilized. For example, a list of “at least one of A, B, and C” includes any of the combinations A, B, C, AB, AC, BC, and/or ABC (i.e., A, B, and C). Furthermore, to the extent more than one occurrence or use of the items A, B, or C is possible, multiple uses of A, B, and/or C may form part of the contemplated combinations. For example, a list of “at least one of A, B, and C” may include AA, AAB, AAA, BB, etc.
(56) While illustrative and presently preferred embodiments of the disclosed systems, methods, and/or machine-readable media have been described in detail herein, it is to be understood that the inventive concepts may be otherwise variously embodied and employed and that the appended claims are intended to be construed to include such variations except as limited by the prior art. While the principles of the disclosure have been provided in connection with specific apparatuses and methods, it is to be clearly understood that this description is made only by way of example and not as a limitation on the scope of the disclosure.

Claims

1. A golf green characteristic indicator for indicating a characteristic of a golf green, the golf green characteristic indicator comprising: a ground track comprising: a proximal end; a distal end oppositely disposed from the proximal end; a first radius of curvature formed at the distal end of the ground track; and a first coupler disposed at the distal end of the ground track; an air track comprising: a proximal end; a distal end oppositely disposed from the proximal end; a second radius of curvature at the proximal end of the air track, wherein the second radius of curvature is equal to the first radius of curvature of the ground track; and a second coupler disposed at the proximal end of the air track; an alignment pair disposed in either the first coupler or the second coupler, the alignment pair comprising: a first alignment feature; and a second alignment feature; a golf ball release mechanism disposed at the distal end of the air track, the golf ball release mechanism comprising: a support plate adjoined to the distal end of the air track, comprising: a left face; a right face; a central hole; a first torsion spring pocket; and a radial pocket; a golf ball cradle comprising: a central shaft passing through the central hole; a second torsion spring pocket; and a pin adjoining the radial pocket; a knob formed about the central shaft; and a torsion spring

biasingly adjoining the support plate and the golf ball cradle; wherein the left face of the support plate adjoining the knob and the right face of the support plate is adjoining the golf ball cradle; a travel configuration wherein: the first coupler of the ground track is detached from the second coupler of the air track; and an indication configuration wherein: the first coupler is interfaced with the second coupler; the first alignment feature is aligned with the second alignment feature; the proximal end of the ground track is configured to adjoin the golf green; and the golf ball release mechanism is actuated to release a golf ball sequentially along the air track and the ground track with a predetermined release speed as the golf ball exits the proximal end of the ground track to indicate the characteristic of the golf green.

2. The golf green characteristic indicator of claim 1, wherein either of the first alignment feature or the second alignment feature comprises: a tapered protrusion.

3. The golf green characteristic indicator of claim 2, wherein either of the first alignment feature or the second alignment feature comprises: a tapered pocket; wherein in the indication configuration the tapered protrusion is slidingly engaged with the tapered pocket.

4. The golf green characteristic indicator of claim 1 and further comprising: a first magnet disposed in either the first coupler or the second coupler.

5. The golf green characteristic indicator of claim 4 and further comprising, in either of the first coupler or the second coupler any one of: a second magnet, or a ferromagnetic ring oppositely formed to the first magnet.

6. The golf green characteristic indicator of claim 1 and further comprising: at least one of a predetermined list of features formed in the proximal end of the ground track, the predetermined list of features comprising: a hook configured to receive a loop of a ruler; a transverse slot configured to receive an end hook of a tape measure; and a plurality of golf tee slots.

7. The golf green characteristic indicator of claim 1 and further comprising: a stand, comprising: a pair of legs; a web coupled to each of the pair of legs, wherein each of the pair of legs is rotationally coupled to the ground track; and wherein in the travel configuration the web is adjoining the ground track; wherein in the indication configuration the web is rotationally decoupled from the ground track by a predefined angle relative to the proximal end of the ground track.

8. The golf green characteristic indicator of claim 7 and further comprising: a recess formed in the ground track at the proximal end to accommodate the web in the travel configuration.

9. The golf green characteristic indicator of claim 1 and further comprising: a continuous curved profile continuously formed in both the ground track and the air track.

10. A golf green characteristic indication method for indicating a characteristic of a golf green, the golf green characteristic indication method comprising: providing a ground track, the ground track comprising: a proximal end; distal end oppositely disposed from the proximal end; a first radius of curvature formed at the distal end of the ground track; and a first coupler disposed at the distal end of the ground track; providing an air track, the air track comprising: a proximal end; distal end oppositely disposed from the proximal end; a second radius of curvature at the proximal end of the air track, wherein the second radius of curvature is equal to the first radius of curvature of the ground track; and a second coupler disposed at the proximal end of the air track; providing an alignment pair in the first coupler and the second coupler, the alignment pair comprising: a first alignment feature; and a second alignment feature; providing a golf ball release mechanism disposed at the distal end of the air track, the golf ball release mechanism comprising: a support plate adjoined to the distal end of the air track, comprising: a left face; a right face: a central hole; a first torsion spring pocket; and a radial pocket; a golf ball cradle comprising: a central shaft passing through the central hole; a second torsion spring pocket; and a pin adjoining the radial pocket; a knob formed about the central shaft; and a torsion spring biasingly adjoining the support plate and the golf ball cradle; wherein the left face of the support plate adjoining the knob and the right face of the support plate is adjoining the golf ball cradle; and configuring the ground track and the air

track between: a travel configuration, comprising: detaching the first coupler of the ground track from the second coupler of the air track; and an indication configuration, comprising: interfacing the first coupler with the second coupler; aligning the first alignment feature with the second alignment feature; the proximal end of the ground track is configured to adjoin the golf green; and actuating the golf ball release mechanism for releasing a golf ball sequentially along the air track and the ground track with a predetermined release speed as the golf ball exits the proximal end of the ground track for indicating the characteristic of the golf green.

11. The golf green characteristic indication method of claim 10, wherein providing the alignment pair further comprises: either of the first alignment feature or the second alignment feature comprising: a tapered protrusion.

12. The golf green characteristic indication method of claim 11, wherein providing the alignment pair further comprises: either of the first alignment feature or the second alignment feature comprising: a tapered pocket; wherein in the indication configuration the tapered protrusion is slidingly engaged with the tapered pocket.

13. The golf green characteristic indication method of claim 10 and further comprising: providing a first magnet, wherein the first magnet disposed in either the first coupler or the second coupler.

14. The golf green characteristic indication method of claim 13 and further comprising: providing, in either of the first coupler or the second coupler any one of: a second magnet, or a ferromagnetic ring oppositely formed to the first magnet.

15. The golf green characteristic indication method of claim 10 and further comprising: providing at least one of a predetermined list of features in the proximal end of the ground track, the predetermined list of features comprising: a hook configured to receive a loop of a ruler; a transverse slot configured to receive an end hook of a tape measure; and a plurality of golf tee slots.

16. The golf green characteristic indication method of claim 10 and further comprising: providing a stand, comprising: a pair of legs; and a web coupled to each of the pair of legs, wherein each of the pair of legs is rotationally coupled to the ground track; wherein in the travel configuration, the web is adjoining the ground track; wherein in the indication configuration, the web is rotationally decoupled from the ground track by a predefined angle relative to the proximal end of the ground track.

17. The golf green characteristic indication method of claim 16 and further comprising: providing a recess in the ground track at the proximal end to accommodate the web in the travel configuration.

18. The golf green characteristic indication method of claim 10 and further comprising: providing a continuous curved profile, wherein the continuous curved profile continuously formed in both the ground track and the air track.